

Assignment: Fashion-MNIST Image Classification Using ANN

Build an **Artificial Neural Network (ANN)** to classify Fashion-MNIST images into 10 clothing categories and develop a simple Streamlit application for prediction.

Course: Artificial Intelligence / Deep Learning

Development Environment: VS Code

Framework: TensorFlow / Keras

Application: Streamlit

Requirements

1. Dataset & Preprocessing

- Use the **Fashion-MNIST dataset**.
- Explore and display sample images.
- Normalize pixel values to **0–1**.
- Convert 28×28 images into **784 input features** for the ANN.
- Display the 10 class names.

2. ANN Model

Build an ANN with:

- At least **2 hidden layers**
- **ReLU** activation for hidden layers
- **10-neuron Softmax** output layer
- Optimizer: **Adam**
- Loss: **Sparse Categorical Crossentropy**
- Metric: **Accuracy**

3. Model Training & Evaluation

- Train the model using a suitable number of epochs.
- Use validation data during training.
- Report:
 - Training accuracy
 - Validation accuracy

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- Test accuracy
 - Test loss
- Plot:
 - Accuracy graph
 - Loss graph
- Create a **confusion matrix**.
- Generate a **classification report**.
- Show examples of correct and incorrect predictions.

4. Model Experiment

Change **at least one** model/training parameter, such as:

- Number of hidden layers
- Number of neurons
- Epochs
- Batch size
- Dropout

Compare the original model with the modified model.

Model	Configuration	Test Accuracy
Original	_____	_____%
Modified	_____	_____%

Briefly explain the result.

5. Streamlit Application

Create a simple Streamlit application that:

- Loads the trained .keras model.
- Accepts an image as input.
- Preprocesses the image.
- Predicts the Fashion-MNIST class.
- Displays the **predicted class and confidence**.

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6. Report

Submit a short report containing:

1. Introduction
2. Dataset & Preprocessing
3. ANN Architecture
4. Training Results
5. Model Evaluation
6. Experiment & Comparison
7. Streamlit Application
8. Conclusion

Submission

Submit the complete project folder containing:

- train_model.py
- app.py
- fashion_mnist_ann.keras
- requirements.txt
- results/
 - accuracy.png
 - loss.png
 - confusion_matrix.png
 - report.pdf

Suggested Filename

StudentName_Fashion_MNIST_ANN.zip

All code must be developed and tested in VS Code.